

Boyuan Chen

Senior Principal Researcher & Technical Lead

LLM Post-Training & Agentic Learning

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Website · Google Scholar · GitHub · LinkedIn

Research and technical leader with industrial-scale LLM post-training experience across thousands of Ascend NPUs. Lead an approximately 20-person team spanning data, environments, SFT/RL, and evaluation. Combine post-training research and distributed-systems engineering, with publications at EMNLP, AACL-IJCNLP, ICSE, and ASE.

Experience

Huawei Canada | Centre for Software Excellence

Markham, ON

Senior Principal Researcher & Technical Lead

May 2026 – Present

Previously: Senior Researcher → Principal Researcher

Dec 2022 – Apr 2026

Industrial-scale post-training. Lead approximately 20 researchers and engineers across Pangu code-model SFT and RL for 8B–718B models, coordinating data curation, environments, NPU training, and agent evaluation.

SFT data curation. Contributed to **MindForge**, an automated SFT data pipeline spanning source-free environments, teacher rollouts, build-validity filtering, infrastructure-failure recovery, and reasoning repair. Fine-tuning Qwen3.6-27B on 973 curated trajectories raised ProgramBench average test pass rate from 37.98% to 49.51%, with gains across seven additional SE benchmarks.

End-to-end agent learning. Led **RepoForge**, integrating repository mining, 7,304 executable environments, teacher trajectories, SFT, and RL. Its 8B agent achieved 17.4% on SWE-bench Verified, leading the $\leq 8B$ non-thinking category at its August 2025 release.

Data-centric capability improvement. Co-authored an industrial study that increased usable teacher supervision $2.84\times$ with the same teacher and attempt budget. The case study improved held-out LiveCodeBench v6 pass@1 by 6.11 percentage points and CodeForces by 2.59 points, while keeping AIME/MATH regression suites within tolerance.

Cross-scaffold generalization. Built trajectory collection, filtering, and distillation pipelines across Claude Code, OpenCode, and OpenHands. Co-authored DCAS, studying planning-aware fine-tuning that improved performance across non-training agent scaffolds.

Data quality. Designed SPICE for issue-clarity, test-coverage, and effort labeling, reporting approximately $19,000\times$ lower cost than estimated manual annotation in a 1,000-instance comparison.

Trustworthy evaluation. Designed SWE agent evaluations across bug fixing, feature implementation, code editing, and architecture; co-authored SWE-Effi and When Elo Lies to examine resource-bounded performance and Codeforces evaluation bias.

Distributed systems and open source. Led heterogeneous-computing research spanning Ascend NPUs and NVIDIA GPUs, including Ray-NPU at 10,000-NPU scale. The team contributed 50+ upstream Ray pull requests to improve cluster scalability, stability, and performance.

Research Intern → Senior Researcher

Apr 2020 – Dec 2022

Researched reproducible deep learning, build systems, and code clones; published in ICSE, TSE, TOSEM, and ICSE-SEIP, including first-author work on training reproducible deep learning models.

Selected Talks & Service

ICSE 2026 technical briefing — Co-presenter, Software Engineering for Foundation Models (SE4FM).

Ray Summit 2024 — Speaker, scaling Ray to 10,000 Huawei Ascend NPUs.

Service: ASE 2024 Program Committee; TSE reviewer (2021–2025).

AI & NLP CONFERENCE PAPERS

LLM Post-Training as Brownfield Maintenance: An Industrial Perspective on Dataware Engineering
EMNLP 2026, Industry Track (accepted) · Co-author
Data-centric post-training under fixed budgets, with targeted capability gains and regression control.

Beyond Tokens: Semantic-Aware Speculative Decoding for Efficient Inference by Probing Internal States
EMNLP 2026, Main Conference · Co-author
Semantic-aware speculative decoding; up to $2.7\times$ speedup on DeepSeekR1-32B.

Beyond Correctness: Enhancing Architectural Reasoning in Code LLMs via Scalable Labeling with Agentic Judgment
AAACL-IJCNLP 2026, Main Conference · Co-author
Architecture-aware data curation and fine-tuning via scalable agentic judgment.

AGENT LEARNING, EVALUATION & REPRODUCIBILITY

MindForge: Teaching Small Language Models Whole-Life-Cycle Software Engineering via Source-Free Program Synthesis
arXiv preprint, 2026 · Co-author
Cited in Xiaomi’s CodeMidas paper; CodeMidas is used to synthesize coding tasks for MiMo-V2.6 RL.

When Elo Lies: Hidden Biases in Codeforces-Based Evaluation of Large Language Models in Practice
ASE 2026, Industry Showcase (accepted) · Co-author

DCAS: Decoupling CLI Agent Scaffolding to Internalize Planning across Scaffolds
ASE 2026, Industry Showcase (accepted) · Co-author

Technical Report: RepoForge: Training a SOTA Fast-thinking SWE Agent with an End-to-End Data Curation Pipeline Synergizing SFT and RL at Scale
arXiv technical report, 2025 · Co-author

Towards Training Reproducible Deep Learning Models
ICSE 2022 · First author

Earlier Experience

Baidu | Cloud Testing Group — Research Intern Aug – Dec 2017
Built a log-based code-coverage estimation prototype evaluated on five industrial projects; published in ASE 2018.

IBM Canada | Platform Symphony — Research Partner Jan – Aug 2016
Built a Hadoop/Python framework to analyze distributed-system logs and detect logging anti-patterns and problematic message sequences.

Education

York University Sep 2014 – Oct 2020
Ph.D., Computer Engineering | NSERC Canada Graduate Scholarship – Doctoral (CGS-D)

University of Science and Technology of China Sep 2010 – Jun 2014
B.E., Computer Science

Technical Focus

Training	SFT; RLVR; multi-turn, execution-feedback RL; teacher-trajectory distillation.
Data, evaluation & systems	Data curation and labeling; agent evaluation; Ray; distributed training; heterogeneous NPU/GPU computing.
Research interests	Domain adaptation, continual learning, and learning from software execution feedback.